

a Held-out reference identities

C. roseus: 256 leaf cells



● Idioblast cell ● Phloem parenchyma
 ● Leaf epidermis ● Unknown
 ● Meristematic cell ● Vascular tissue
 ● Mesophyll

b Strict zero-shot outputs

Target labels never enter fitting or inner-fold selection

**c Primary denominator remains intact**

A coverage-aware report keeps open-set cells visible

all test cells

3,964

source-label
covered

2,216

open-set or
unavailable

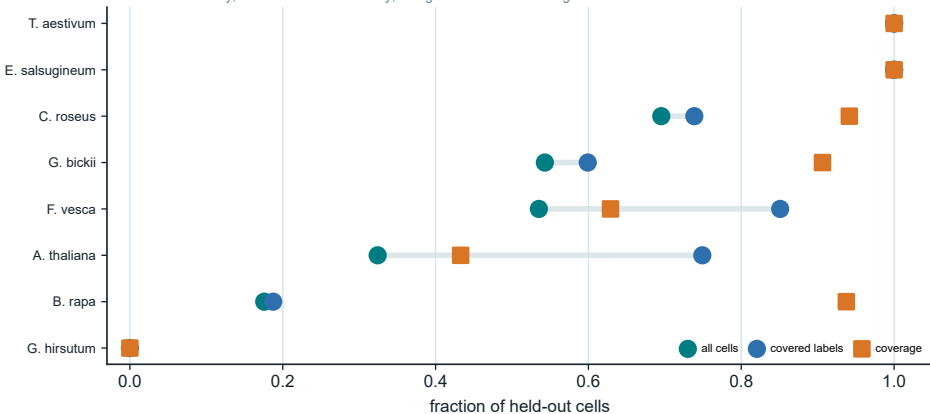
1,748

v17 all-cell accuracy 39.96%

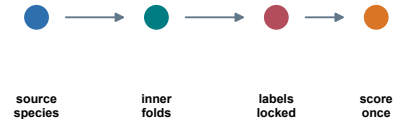
bootstrap central 95% interval 0.384-0.415

d Strict transfer is heterogeneous across all eight held-out species

Teal: all-cell accuracy; blue: conditional accuracy; orange: source-label coverage

**e Nested selection firewall**

A local rule is selected inside source-species folds



No target-cell identity informs candidate choice, thresholding or calibration.